

Udara Roga

Learning Objectives

- ⇒ To learn the concept of Udara
- ⇒ To know about the Pratyatma Lakshana of Udara
- ⇒ To learn the Nidana and Samprapti of Udara mentioned in Ayurveda literatures
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of disease
- ⇒ To enlist the types of Udara
- ⇒ To know about the Purvarupa, Samanya Lakshana, Sadhyasadyata of Udara
- ⇒ To know the Avastha Bheda of Udara
- ⇒ To know the concept of Doshaja Udara
- ⇒ To know the specific Lakshana of Baddhagudodara, Plihodara, Jalodara, Chidrodara

Introduction



Figure 74: Pratyatma Lakshana of Udara

In literal sense, the term *Udara* refers to the anatomical structure i.e. abdomen. In the context of *Roganidana*, *Udara* is the spectrum of diseases, where in we can observe the generalized swelling of the abdominal region. This is different from *Gulma* which is usually a localised swelling in the abdominal region. *Udara* is due to the accumulation of fluids and *Dosha* in the abdominal region. When you

speak about the part of the body as *Udara* which extends from the largest muscle of the body, diaphragm till the pelvis. Diaphragm separates abdomen from thorax. In the current understanding, generalized abdominal swelling arises in case of obesity, ascites, chronic constipation, chronic flatus accumulation and fetus in case of pregnant women. This is *Kapha* and *Vata* dominant disorder with the affliction of *Rasa*, *Udaka* and *Sveda*. In this disease, affliction of *Rasavaha*, *Udakavaha*, *Svedavaha*, *Pranavaha*, *Annava* and *Purishavaha* Srotas is seen with *Sanga* and *Vimargagamana* as the type of Srotodushti occurring during the progression of the disease.

Among the classification of *Vikara* as per *Sushruta*, this is categorized under *Adhyatmika Vikara*, further as *Amashayasamuttha Vikara* and *Sharirika Vikara* as further subcategory of the disease.

उदराण्युत्सेधसामान्यात् शोथप्रभेदा एव॥

Chakra, Ca.Su. – 19/3

Chakrapani says regarding *Udara* as the variety of *Shotha* only because of common feature of swelling.

Pratyatma Lakshana of Udara

उदरोत्सेधसाधर्म्यादुदर

Madhu, Ma.Ni. – 35/1

Pratyatma Lakshana (cardinal feature) of *Udara* is

generalized swelling of abdominal region. part of the body present between xiphisternum to pubic symphysis. This is different from Gulma as the disease will have localized swelling in the abdominal region.

Nidana of Udara

रोगाः सर्वेऽपि मन्देऽग्नौ सुतरामुदराणि च।
अजीर्णान्मलिनैश्चात्रैर्जायन्ते मलसंचयात्॥ 1 ॥
अत्युष्णालवणक्षारविदाखम्लगराशनात्॥
मिथ्यासंसर्जनादूक्षविरुद्धाशुचिभोजनात्।
श्रीहाशौग्रहणीदोषकर्शनात् कर्मविश्रमात्॥ 2 ॥
क्लिष्टानामप्रतीकाराद्रीक्ष्याद्वेगविधारणात्॥ 3 ॥
स्रोतसां दूषणादामात् संक्षोभादति पूरणात्।
अशीबालशरुद्रोधादन्प्रस्फुटन मेदनात्॥ 4 ॥
अतिसंचितदोषाणां पापं कर्म च कुर्वताम्।
उदाप्युपजायन्ते मन्दाग्नीनां विशेषतः॥ 5 ॥

Ma.Ni. – 35/1

Ca.Ci. – 13/12-15

- When the person is suffering from *Mandagni*¹ (reduced digestive fire).

- When an individual is suffering from *Ajirna*² (indigestion).
- When the person consumes contaminated food substances³ excessively.
- Excess consumption of *Ushna Virya* (hot in potency) substances.
- Excess consumption of *Lavana*, (salty), *Amla* (sour) *Rasa* substances.
- Excess consumption of alkaline substances.
- Excess consumption of *Vidahi* (producing burning sensation and thirst) substances⁴.
- Exposure to *Gara Visha*.
- Improper post *Shodhana* measures⁵.
- Excess consumption of *Ruksha* substances⁶.
- Excess consumption of unwholesome food substances⁷.
- When an individual is suffering from other diseases such as *Pliha*, *Arshas*, *Grahani*, emaciation of the body⁸.
- Improper *Shodhana* therapy⁹.

- In this context, the term Mandagni can be considered as poor secretion of hydrochloric acid. When there is reduced secretion of hydrochloric acid, where the pH of the gastric juice above 5, there will be no proteolytic activity takes place because of the inactivation of pepsinogen into pepsin. This will lead to poor protein absorption into the body. Thus, the person may develop the state of hypoproteinemia and this may lead to the generalised swelling of the abdomen or whole body itself.
- In the present context, the person will have either poor hydrochloric acid activity or inactivated pepsinogen. Due to this, the person may have poor absorption of protein resulting in the state of hypoproteinemia which further may lead to excess accumulation of fluid in the interstitial compartment.
- When an individual consumes contaminated food substances, inflammation of the stomach and intestines will be the first mechanism to occur, which further leads to poor absorption of proteins and electrolytes such as sodium, potassium, calcium ions. This will lead to the reduction in oncotic pressure which in turn may lead to the shift of fluid from intravascular compartment to the interstitial compartment and if the accumulation of fluid happens in the peritoneal cavity results in ascites.
- When an individual is having the exposure to these kinds of aetiological factors, he may have the vitiation of Pitta Dosha in the body. As per current understanding, the person who consumes more amount of Ushna substances, Lavana and Amla Rasa substances and Vidahi substances continuously, that may lead to inhibition to the gastric emptying as highly irritating substances coming in contact with duodenal mucosa lead to the closure of pyloric antrum. At the beginning of pathological events, this is termed as Alasaka. In due course of time, if an individual is involved in such kind of activities, that may be responsible for abdominal swelling, stomach in particular. These are the probable aetiological factors for the manifestation of malignant swelling of the stomach and other organs of alimentary canal.
- Let us take an example of consumption of curds soon after the post Shodhana measures, that will lead to the sudden vitiation of either Pitta or Kapha Dosha in the body.
- When the person consumes excess Ruksha (dry) substances, that will lead to the increase of Ruksha quality in the body, which leads to the vitiation of Vata Dosha further. If the person continuously consumes these substances, there will be decrease of Snigdha (unctuous) quality in the alimentary canal. It is being said that poor intestinal motility in the transverse colon, there will be greater absorption of water from the faecal matter and hardness of the faecal matter develops. When there is more dryness present in the chyme, that will lead to excess transport of water into the chyme into the small intestine. This mechanism will lead to the reduced oncotic pressure in the portal veins thus it may initiate the pathological event of ascites.
- In this context, there will similar pathological mechanism which is described in the context of contaminated food substances intake.
- When an individual is suffering from Pliha (any splenic disorder), the person may have ascites. This is attributable to cirrhosis of liver leading to portal hypertension. when there is backflow of blood occurring in superior mesenteric vein, the person may present with bleeding per rectum where in the person will be suffering from internal haemorrhoids. This is the Lakshana seen in Arshas. In case of Grahani, the person may have abdominal swelling attributable to marked nutrition deficiency, proteins in particular. In case of Grahani, the person may have generalised oedema.
- As per Ayurveda literatures, the term Shodhana refers to purificatory procedures Vamana (emesis), Virechana (purgation), Basti (medicated enema), Nasya (nasal administration of medicine) and Raktamokshana (blood letting). The term improper Shodhana therapy refers to excessive or inadequate Shodhana therapy. Both of these mechanisms may lead to two types of events either there may be non progressive (compensated) shock or progressive shock.

Non progressive (compensated) shock:

- Baroreceptors reflexes which elicit powerful sympathetic stimulation of the circulation
- Central nervous system ischemic response, occurs when arterial pressure falls below 50 mmHg

Not following any kind of treatment in serious diseases¹⁰.

Suppression of natural urges¹¹.

In affliction of Srotas¹².

When an individual is having *Ama* in the body¹³.

When an individual is consuming excessively irritating substances^{14,15}.

When an individual consumes increased amount of nutritive substances.

3. Reverse stress-relaxation of the circulatory system which causes the blood vessels to contract around the diminished blood volume, so that the blood volume is available more adequately fills the circulation.
4. Formation of angiotensin by the kidney, which constricts peripheral arteries and causes decreased output of water and salt by the kidneys.
5. Formation of vasopressin (antidiuretic hormone) by posterior pituitary gland which constricts peripheral arteries and veins and greatly increases water retention by the kidneys.
6. Compensatory mechanisms that return the blood volume back toward normal, including absorption of large quantities of fluid from the intestinal mucosa and absorption of fluid into the blood capillaries from the interstitial compartment.

Progressive shock: Following events may happen in case of progressive shock

1. Cardiac depression: Coronary blood flow decreases below that required for adequate nutrition of myocardium.
2. Vasomotor failure: This occurs because of poor blood supply to the brain, if the arterial pressure falls below 30 mmHg in particular.
3. Blockage of very small blood vessels: Because of poor blood flow along the small blood vessels, there will be sluggish movement of blood which is termed as sludged blood. Further, there will accumulation of carbonic and lactic acid along with secretion of other deterioration products from the ischemic tissues may block the movement of blood along the microvessels.
4. Increased capillary permeability: happens because of capillary hypoxia leading to the capillary damage.
5. Generalised cellular deterioration: This is seen in liver, lungs and heart.
 - Active transport of sodium and potassium ions is greatly diminished. This will lead to excess accumulation of sodium and chloride ions inside the cells and extensive movement of potassium ions outside the cells.
 - Mitochondrial activity of the liver cells and other tissues becomes severely depressed
 - Lysosomes in the cells in widespread tissue areas begin to open, with intracellular release of hydrolases which further leads to intracellular degeneration.
 - Cellular metabolism of nutrients, such as glucose becomes greatly depressed.

Thus, improper Shodhana therapy may depress the hepatic function. Thus in the long run, the person may develop ascites.

If an individual is suffering from serious illness and he is not following any of the treatment, the person may have nutrition deficiency, carbohydrates and electrolytes in particular. This will lead to fat mobilisation into the liver and this event may be a prequel to chronic liver disease, which may manifest as a cirrhosis of liver. Cirrhosis of the liver is one of the prequel of manifestation of ascites.

Let us take the example of suppression of faecal matter, that will lead to the vitiation of Apana Vata which in turn results in the reverse movement of Apana Vata which further leads to the poor movements of the intestine leading to the distension of the abdomen. In this occasion, the person will have reverse movement of Samana Vata, which further leads to the generalised distention of the abdomen. Later, the person may have reverse movement of Prana Vata.

Let us take the example of Raktavaha Srotas affliction. In case of affliction of this Srotas, Yakrit or/and Pliha may be afflicted in due course of time depending upon the entry of pathogen into the body, virulence of pathogen in the body etc. When Yakrit (liver) is afflicted, that may lead to the damage of the liver, which may lead to the cirrhosis of the liver, which further leads to the portal hypertension and the outcome of which is manifestation of ascites.

It is being said that normally functioning liver performs around 72 different physiological functions of the body. Depending upon the nature of the aetiological factor exposure, the person may have compromise in few or all physiological functions. Thus, the person may have chronic liver disease. This may be either in the form of enlargement of liver or cirrhosis of liver. Thus, the person may have Yakritodara.

During portal hypertension, the person may have enlargement of spleen secondary to backflow of blood in the splenic vein via backflow of the blood in portal vein. This may be attributable to Plihodara.

When there is *Ama* in the body, it may be seen either in the alimentary canal or in the tissues level. When there is *Ama* in the body, it may be responsible for the reduced nutrient formation and assimilation to the body.

When nutrient deficiency is present in the body, this may lead to the sequential depletion of carbohydrates, fat and proteins. This may lead to the manifestation of metabolic acidosis (either ketoacidosis due to increased fat metabolism or lactic acidosis due to increased muscle nutrient metabolism).

When fat deposited in the adipose tissue is going to get mobilised, it may move into the liver forming the state of fatty degeneration of liver. This is earlier phase of cirrhosis of liver. This may lead to portal hypertension and thus the person may have the ascites manifestation.

In this context, the pathological mechanism occurs similar to the description described in the context of excessive contaminated substances intake.

When the person is consuming excess intake of irritating substances, this may lead to the inflammation of the alimentary canal, this may lead to nutrient depletion in the body, sequential loss of carbohydrate, fats and proteins. During fat metabolism, the person may have fatty degeneration of liver, thus this may lead to portal hypertension as a consequence of cirrhosis of liver which may develop after fatty degeneration of liver.

- When an individual is suffering from constipation^{16,17}
- When trauma occurs to the abdomen¹⁸
- When there is excess accumulation of *Dosha* in the body
- When an individual is doing *Papa Karma* (socially banned activities)

Samprapti of Udara

रूमा स्वेदाम्बुवाहीनि दोषाः स्रोतांसि संचिताः।

प्राणाशयपानान् संदृष्ट्य जनयन्त्युदरं नृणाम्॥20॥ Ca.Ci. – 13/20

- Excessively accumulated *Dosha* will obstruct the *Sveda* and *Udakavaha Srotas*, which will lead to displacement of *Udaka* to the *Udara* region
- In this disease, there will be affliction of *Pranavata*, *Pachakagni* and *Apanavata*.

Samprapti Ghataka

- Dosha** : *Kapha, Vata*
- Dushya** : *Rasa, Udaka, Sveda*
- Srotas** : *Rasavaha, Udakavaha, Svedavaha, Annavaha, Pranavaha Srotas*
- Srotodushti Prakara**: *Sanga and Vimargagamana*
- Agni** : *Jatharagnimandya*
- Ama** : *Jatharagnimandyaajanya*
- Udbhava Sthana** : *Amashaya*
- Sanchara Sthana** : *Sarva Sharira*
- Vyakta Sthana** : *Udara*

- Rogamarga** : *Abhyantara, Bahya and Madhyama*
- Vyadhi Svabhava** : *Chirakari*
- Sadhyasadhyata** : *Asadhyata*

Purvarupa of Udara

क्षुन्नाशः स्वाद्वतिस्निग्धगुर्वन्नं पच्यते चिरात्।
भुक्तं विदह्यते सर्वं जीर्णाजीर्णं न वेत्ति च॥16॥
सहते नातिसौहित्यमीषच्छोपश्च पादयोः।
शश्वद्वलक्षयोऽल्पेऽपि व्यायामे श्वासमृच्छति॥17॥
वृद्धिः पुरीषेनिचयो रुक्षोदावर्तहेतुका।
बस्ति सन्धौ रुगाध्मानं वर्धते पाठ्यतेऽपि च॥18॥
आतन्यते च जठरमैपि लघ्वल्प भोजनात्।
राजीजन्म बलीनाश इति लिङ्गं भविष्यताम्॥19॥ Ca.Ci. – 13/16-19
तत्पूर्वरूपं बलवर्णकाला बलीविनाशो जठरे हि राज्यः॥17॥
जीर्णापरिज्ञानविदाहवत्यो बस्तौ रुजः पादगतश्च शोफः।

Su.Ni. – 7/7,8

- Loss of appetite¹⁹
- Delayed digestion of sweet, unctuous and heavy substances²⁰
- Improper digestion of ingested food²¹
- The person is not in a position to know the digestion and indigestion of food²²
- The person is not in a position to tolerate slight exertion
- Slight swelling of both feet (bilateral pedal oedema)²³
- Rapid loss of strength²⁴

G
K

- When an individual is suffering from constipation, the first and foremost, there will be affliction of *Apanavata*. This will lead to *Udavarta*, where there will be reverse movement of *Vata* leading to the affliction of *Samanavata* and *Pranavata* respectively in due course of time. When *Samanavata* is afflicted, there will be affliction of *Ushma*, *Kleda* and *Sneha* etc. This will lead to the generalised abdominal swelling and this swelling is attributable to the mechanism of meteorism. Here, the person may have borborygmi on auscultation of the abdomen.
- When a person is suffering from constipation, this may lead to the vitiation and reverse movement of *Apana Vata*, which may lead to the vitiation and reverse movement of *Samana Vata* and at last, this may lead to the vitiation and reverse movement of *Pranavata*. Chronic constipation may lead to generalised abdominal swelling secondary to accumulation of faecal matter.
- Mechanism of the injury may give the clue to the diagnosis. Closed injuries are due to waves of shock or direct compression of a viscus against a bony prominence. If a large segment of abdomen or abdominothoracic wall is compressed, it may burst or split organs like liver and spleen. If similar force is applied in case of breath holding, it may lead to rupture of diaphragm. The structures e.g. bladder, urethra which are attached to bones by fascial bands may be torn if such bones are fractured. In case of penetrating wounds, the length of the weapon and the velocity with which it was struck are very important. High velocity injuries produced by gunshot or fragments from exploding mines and shells penetrate deeply and may damage extensively anything in or around their paths. As far as physical examination of the patient in such occasion is concerned, the patient may have silent abdomen owing to peritonitis or he may have restlessness in case of intraperitoneal haemorrhage. The person may show signs of hypovolemia and anemia. The person may have thoracic movements in case of abdominal injuries. There will be generalised abdominal swelling secondary to meteorism or internal haemorrhage.
- This can be attributable to backward flow of blood occurs in case of portal hypertension which in turn leads to increased secretion in the gastrointestinal lumen.
- As per Ayurveda understanding, the consumption of *Madhura*, *Snigdha* and *Guru* substances will increase the *Kapha Dosha* in the body, thereby hampering the process of digestion.
- This can be attributable to increased secretion observed in the gastrointestinal lumen.
- This can be attributable to increased secretion in the alimentary canal because of backflow of the blood occurs as a result of portal hypertension.
- This can be attributable to hypoproteinemia where reduced amount of proteins in the blood is seen.
- This can be attributable to reduced absorption of water and electrolytes – sodium, potassium, calcium etc. For example, sodium ions can be rapidly depleted in case of gastrointestinal dysfunction. The outcome of this event is very rapid so that the serious symptoms are seen within hours.

- Breathlessness on performing slight activities²⁵
- Increased accumulation of faecal matter
- Pain on either sides of iliac region
- Pain in the lower abdomen
- Distension of the abdomen
- Fullness of abdomen on consuming slight amount of food which is even light to digest²⁶
- Appearance of veins in the abdomen²⁷
- Loss of wrinkles in the abdomen or shininess of abdomen²⁸

Samanya Lakshana of Udara

कुक्षेराध्मानमाटोपः शोफः पादकरस्य च।
मन्दोऽग्निः श्लक्ष्णगण्डत्वं कार्ये चोदरलक्षणम् 21 Ca.Ci. – 13/21
आध्मानं गमनेऽशक्तिर्दौर्बल्यं दुर्बलाग्निता।
शोथः सदनमङ्गानां सङ्गो वातपुरीषयोः॥3॥
दाहस्तन्ना च सर्वेषु जठरेषु भवन्ति हि। Ma.Ni. – 35/3-4

- Distension of the abdomen²⁹
- Gurgling sounds in the abdomen³⁰
- Swelling of the upper and lower limbs³¹
- Loss of appetite³²
- Sliminess of the temples
- Emaciation of the body³³
- Difficulty in moving
- Debility
- Sluggish movement of the body parts³⁴
- Constipation
- Retention of urine³⁵

Bheda of Udara

पृथग्दोषैः समस्तैश्च प्लीहबद्धक्षतौदकैः।

- This can be attributable to push of the diaphragm upwards, so that compression of the thorax occurs leading to occurrence of breathlessness.
- As per Ayurveda literatures, because of above said reasons, first of all there will be vitiation of Apanavata, which further leads to the reverse movement of Vata leading to the affliction of Samanavata and at the end, there will be the affliction of Pranavata in due course of time if proper management of the disease is not met.
- This can be attributable to the development of collaterals of blood vessels so that blood supply to the extra tissue growth can be met.
- This can be attributable to stretching of the abdominal wall and thinning of the abdominal wall.
- This can be attributable to stretching of the abdominal wall and thinning of the abdominal wall.
- This is a globular shaped abdomen. This can occur when there is excess accumulation of faeces, flatus or fluid in the abdominal region.
- The presence of gurgling sounds in the abdomen may indicate poor intestinal motility which can be seen in partial intestinal obstruction.
- The presence of gurgling sounds in the abdomen may indicate poor intestinal motility which can be seen in partial intestinal obstruction.
- This can also be observed in early phase of Udara disease.
- Swelling of the upper and lower limbs seen in case of protein deficiency in the body which is observed in the patients of chronic liver disease.
- This can be attributable to backflow of the blood into alimentary canal.
- This can be attributable to the increased fat loss from the body. This symptom occurs because of poor nutrient absorption from the body.
- Fat depletion occurs from buttocks, breasts, subcutaneous tissue and abdomen.
- This can be attributable to hampered absorption of nutrients including water, sodium, potassium, calcium ions which are essential for all kinds of skeletal, smooth and cardiac muscular activity and nerve conduction. Thus, poor absorption of these nutrients leads to sluggish movement of body parts.
- Constipation and retention of urine can be attributable to vitiation of Apanavata which further leads to sequential affliction of Samanavata and Pranavata.

संभवन्त्युदराण्यष्टौ तेषां लिङ्गं पृथक् शृणु॥2२॥ Ca.Ci. – 13/22

8 types of Udara as per Charaka

1. Vataja Udara
2. Pittaja Udara
3. Kaphaja Udara
4. Sannipataja Udara
5. Plihodara
6. Baddhodara
7. Kshatodara
8. Udakodara or Jalodara

पृथक्समस्तैरपि चेह दोषैः लीहोदरं बद्धगुदं तथैव।

आगन्तुकं सप्तममष्टमं चन्द्रं चेति वदन्ति तानि॥4॥ Su.Ni. – 7/4

8 types of Udara as per Sushruta

1. Vataja Udara
2. Pittaja Udara
3. Kaphaja Udara
4. Sannipataja Udara
5. Plihodara
6. Baddha Gudodara
7. Agantuja (Parisravi) Udara
8. Jalodara

Vataja Udara

रूक्षाल्पभोजनायासवेगोदावर्तकर्शनैः।

वायुः प्रकुपितः कुक्षिहृद्वस्तिगुदमार्गः॥2३॥

हत्वा कफमुद्ध्य तेन रुद्धगतिस्ततः।

आचिनोत्युदरं जन्तोस्त्वङ्मांसान्तरमाश्रितः॥2४॥

तस्य रूपाणि कुक्षिपाणिपादवृषणश्चयथुः, उदरविपाटनम्,

अनियतौ च वृद्धिहासौ, कुक्षिपार्श्वशूलोदावर्ताङ्गमर्दं पर्वं भेदशुष्ककास-

कार्यदौर्बल्यारोचकाविपाकाः, अधोगुरुत्वं, वातवर्चोमूत्रसङ्गः, श्यावा-
रुणत्वं च नखनयनवदनत्वमूत्रवर्चसाम्, अपि चोदरं तन्वसितराजी-
सिरासंततम्, आहतमाध्मातहतशब्दवद्भवति, वायुश्चोर्ध्वमधस्तिर्यक्
च सशूलशब्दश्चरति, एतद्वातोदरमिति विद्यात् 25

Ca.Ci. - 13/23-25

- ⊗ Consumption of dry substances
- ⊗ Consumption of very little amount of food substances
- ⊗ Excess exertional activities
- ⊗ Suppression of natural urges
- ⊗ Measures leading to emaciation of the body
- ⊗ First, there will be Agnimandya which will be followed by vitiation of Kapha Dosha leading to the obstruction of movement of Vata
- ⊗ Accumulation of Jala in abdominal region

Lakshana of Vataja Udara

- ⊗ Swelling of the abdomen, upper limbs, lower limbs, scrotum
- ⊗ Excess stretch of the abdominal region
- ⊗ Disproportionate increase and decrease of abdominal girth
- ⊗ Pricking type of pain in stomach and sides of the abdomen
- ⊗ Upward movement of obstructed Vata from Pakvashaya
- ⊗ Bodyache
- ⊗ Pain in the joints
- ⊗ Dry cough
- ⊗ Emaciation
- ⊗ Debility
- ⊗ Tastelessness
- ⊗ Impaired digestion
- ⊗ Heaviness of the lower part of the body
- ⊗ Constipation, retention of urine and suppression of flatus
- ⊗ Blackish or reddish discolouration of nails, eyes, face, urine and faecal matter
- ⊗ Abdominal swelling filled with thin varicose veins
- ⊗ Succussion splash can be heard
- ⊗ Upward, downward and oblique movement of vitiated Vata Dosha associated with pricking type of pain and sounds

Pittaja Udara

कट्वम्ललवणात्युष्णतीक्ष्णान्यातपसेवनैः।
विदाह्यशनाजीर्णैश्चाशु पित्तं समाचितम्॥26॥
प्राप्यानिलकफौ रुवा मार्गं मुन्मार्गमास्थितम्।
निहन्त्यामाशये बहिं जनयत्युदरं ततः॥27॥

रूपाणि- दाहज्वरतृष्णामूर्च्छातीसार- भ्रमाः, कटुकास्यत्वं,
हरितहारिद्रत्वं च नखनयनव- दनत्वङ्मूत्रवर्चसाम्, अपि चोदरं नील
पीतहारिद्र- हरितताम्रराजीसिरावनद्धं दह्यते, दूयते, धूयते, ऊष्मायते,
विद्यते, विकट्यते, मृदुस्पर्शं क्षिप्रपार्कं च भवति; एतत् पित्तोदरमिति
विद्यात्॥28॥

Ca.Ci. - 13/26-28

- ⊗ Excess consumption of Katu (pungent), Amla (sour) and Lavana (salt) Rasa substances
- ⊗ Excess consumption of Ushna Virya (hot in potency) substances
- ⊗ Excess consumption of highly irritating substances
- ⊗ Excess exposure to fire and sunlight for example, cooks, manual labourers
- ⊗ Excess consumption of Vidahi (producing burning sensation in the chest and thirst) food substances
- ⊗ Consumption of food before the digestion of previously ingested meal
- ⊗ First there will be vitiation of Pitta associated with Vata and Kapha vitiation and there will be loss of appetite

Lakshana of Pittaja Udara

- ⊗ Burning sensation
- ⊗ Fever
- ⊗ Thirst
- ⊗ Altered state of consciousness
- ⊗ Diarrhoea
- ⊗ Giddiness
- ⊗ Pungent taste in the mouth
- ⊗ Greenish or yellowish discolouration of nails, eyes, face, skin, urine and faecal matter
- ⊗ Abdomen is filled with visible veins which have bluish, yellowish, turmeric coloured, greenish, coppery colour
- ⊗ Lakshana or clinical features indicate the increase of Pitta in the body such as burning sensation, abnormal hot perception, sweating etc
- ⊗ Rapid suppuration

Kaphaja Udara

अव्यायामदिवास्वप्नस्वाद्वतिस्निग्धपिच्छिलैः।
दधिदुग्धोदकानूपमांसैश्चाप्यतिसेवितैः॥29॥
कुब्धेन श्लेष्मणा स्रोतः स्वावृतेष्वावृतोऽनिलः।
तमेव पीडयन् कुर्यादुदरं बहिरन्त्रः॥30॥

तस्य रूपाणि गौरवारोचकाविपाकाङ्गमर्दाः, सुप्तिः,
पाणिपादमुष्कोरुशोफः, उत्क्लेशनिद्राकास - श्वासाः, शुक्लत्वं च
नखनयनवदनत्वमूत्रवर्चसाम् अपि चोदरं शुक्लराजीसिरासंततं, गुरु,
स्तिमितं, स्थिरं, कठिनं च भवति एतच्छ्लेष्मोदरमिति विद्यात्॥31॥

Ca.Ci. - 13/29-31

- Physical inactivity
- Excess sleep during day time
- Excess consumption of unctuous and slimy substances
- Excess consumption of curds, milk and milk products
- Excess consumption of domestic animals and animals of marshy places
- Kapha Dosha will vitiated excessively leading to the obstruction of movement of Vata

Lakshana of Kaphaja Udara

- Heaviness of the abdomen
- Tastelessness
- Impaired digestion
- Bodyache
- Tingling sensation
- Swelling of upper and lower limbs and genital region
- Nausea
- Excess sleep
- Cough
- Breathing discomfort
- Whitish discolouration of nails, eyes, face, skin, urine and faecal matter
- Abdomen is filled whitish coloured varicose veins
- Heaviness of abdomen and body parts
- Sensation of covering of the body with wet cloth
- Firm and hard swelling

Sannipataja Udara

दुर्बलाग्नेरपथ्यामविरोधिगुरुभोजनैः।
स्त्रीदत्तैश्च रजोरोमविण्मूत्रास्थिनखादिभिः॥३२॥
विषैश्च मन्दैर्वाताद्याः कुपिताः संचयं त्रयः।
शनैः कोष्ठे प्रकुर्वन्तो जनयन्त्युदरं नृणाम्॥३३॥

तस्य रूपाणि - सर्वेषामेव दोषाणां समस्तानि लिङ्गान्युपलभ्यन्ते,
वर्णाश्च सर्वे नखादिषु, उदरमपि नानावर्णराजीसिरासंततं भवति; एतत्
सन्निपातो- दरमिति विद्यात्॥३४॥

Ca.Ci. - 13/32-34

- When the person is having poor Agni (digestive fire)
- When the person consumes *Apathya* (unhealthy food), *Ama* (improperly cooked), *Virodhi* (unwholesome food), *Guru* (heavy to digest) food substances
- If the food is contaminated with menstrual blood, body hairs, faecal matter, urine, bone and nails, toxic substances

³⁶ In this type of Udara, enlargement of abdominal region is observed in the left upper quadrant. The nature of the swelling is similar to the back of tortoise.

As per Charaka, 2 types of Pliha Vriddhi can be observed - Chyuta (Visceroptosis of spleen) and Achyuta (no descent of spleen). Sankshobha etc will lead to Chyuta type of Pliha Roga is seen and in case of Rakta Dhatu involvement, Achyuta type of Pliha enlargement is seen.

- Vitiation of all three *Dosha* occurs and responsible for the manifestation of *Udara*.

Lakshana of Sannipataja Udara

- Clinical features of all three *Doshaja* varieties of *Udara*
- Various colours in skin, blood vessels, nails, eyes
- Stretch marks of different colours
- Enlarged abdomen

Plihodara³⁶

अशितस्यातिसंक्षोभाद्यानयानाति चेष्टितैः।
अतिव्यवायभाराध्वमनव्याधिकर्शनैः॥३५॥
वामपार्श्वाश्रितः प्लीहा च्युतः स्थानात् प्रवर्धते।
शोणितं वा रसादिभ्यो विवृद्धं तं विवर्धयेत्॥३६॥
तस्य प्लीहा कठिनोऽष्टीलेवादौ वर्धमानः कच्छपसंस्थान उपलभ्यते स
चोपेक्षितः क्रमेण कुक्षिं जठरमध्यधिष्ठानं च परिक्षिपन्नुदरमभिनिर्वर्त-
यति॥३७॥
तस्य रूपाणि - दौर्बल्यारोचकाविपाकवर्चीमूत्रग्रहतमः प्रवेशपिपा-
साङ्गमर्दच्छर्दिमूर्च्छाङ्गसादकासश्वासमृदुज्वरानाहाग्निनाशकार्यास्यवै-
रस्यप- वैभेदकोष्ठवातशूलानि अपि चोदरमरुणवर्णं विवर्णं वा नील
हरितहारिद्रराजिमद्भवति; एवमेव यदपि दक्षिणपार्श्वस्थं कुर्यात्,
तुल्यहेतुलिङ्गौषधत्वात्तस्य लीहजठर एवावरोध इति एतत् प्लीहोदरमिति
विद्यात्॥३८॥

Ca.Ci. - 13/35-38

- Excess consumption of food immediately after vigorous exercise or travelling or lifting heavy weight etc
- Excess sexual intercourse
- Excessive *Vamana Karma* (emesis)
- As a effect of several primary diseases
- Due to the vitiation of *Rasa*, *Rakta*, there will be increase of *Pliha* which is present in the left side of the abdomen and this will be displaced from its own site. The swelling arising from *Pliha* will slowly enlarge and becomes hard and the swelling simulates the shell of tortoise leading to the increase in size of abdomen and covers the entire abdomen.

Lakshana of Plihodara

- Debility
- Tastelessness
- Impaired digestion
- Constipation
- Retention of urine
- Darkness in front of the eyes

- Thirst
- Bodyache
- Vomiting
- Altered state of consciousness
- Sluggish movement of body parts
- Cough
- Breathing discomfort
- Mild fever
- Distension of the abdomen
- Loss of appetite
- Emaciation
- Abnormal perception of the taste
- Joint pain
- Pain abdomen
- Reddish discolouration of abdomen with bluish or yellowish colour of blood vessels over the abdomen

Baddha Gudodara³⁷

पक्ष्मबालैः सहात्रेन भुक्तैर्बद्धायने गुदे।

उदावर्तैस्तथाऽशभिरन्त्रं संमूर्च्छनेन वा॥३९॥

अपानो मार्गसरोधाद्धत्वाऽग्निं कुपितोऽनिलः।

वर्चः पित्तकफान् रुद्धा जनयत्युदरं ततः॥४०॥

तस्य रूपाणि - तृष्णादाहज्वरमुखतालुशोषोरुसादकासश्वास
दौर्बल्यारोचका विपाकवर्चोमूत्रझाध्मानच्छर्दिक्षवथुशिरोहन्नाभिगुद-
शूलानि, अपि चोदरं मूढवातं स्थिरमरुणं नीलराजि सिरावनद्ध- राजकं
वा प्रायो नाभ्युपरि गोपुच्छवदभिनिर्वर्तत इति एतद्वद्गुदोदरमिति
विद्यात्॥४१॥

Ca.Ci. - 13/39-41

- If the person consumes food substances contaminated with hairs
- If the person is suffering from *Udavarta* (condition where reverse movement of *Vata* is seen) or *Arshas* (hemorrhoids)
- Because of above reasons, the contents of intestines, foreign bodies and faeces gets blocked in the intestine. This is leading to the vitiation of *Apana Vata* due to which there will be reverse movement of *Vata* leading to distension of abdomen.

Lakshana of Baddhagudodara

- Thirst
- Burning sensation
- Fever
- Dryness of mouth, palate
- Sluggish movement of thighs

- Cough
- Breathing discomfort
- Debility
- Tastelessness
- Impaired digestion
- Constipation
- Retention of urine
- Distension of abdomen
- Vomiting
- Sneezing
- Headache
- Chest discomfort
- Pain in the umbilical region
- Pain around anal region
- Abnormal movement of *Vata* in abnormal directions
- Firm and stable swelling of the abdomen
- Abdominal swelling associated with reddish or bluish discolouration of the veins over the abdomen
- The swelling situated above the umbilicus in the shape resembling the tail of cow (broad above and tapering below)

Chidrodara³⁸

शर्करातृणकाष्ठास्थिकण्टकैरन्नसंयुतैः।

भिद्येतात्रं यदा भुक्तैर्नृम्भयाऽत्यशनेन वा॥४२॥

पाकं गच्छेद्रसस्तेभ्यश्छिद्रेभ्यः प्रस्त्रवद्बहिः।

पूरयन् गुदमन्त्रं च जनयत्युदरं ततः॥४३॥

तस्य रूपाणि - तदधो नाभ्याः प्रायोऽभिवर्धमा- नमुदकोदरं भवति,
यथाबलं च दोषाणां रूपाणि दर्शयति, अपि चातुरः सलोहितनील
पीतपिच्छिल- कुणपगन्ध्यामवर्च उपवेशते, हिक्काश्वासकासतृ-
ष्णाप्रमेहारोचकाविपाकदौर्बल्यपरीतश्च भवति; एतच्छिद्रोदरमिति
विद्यात्॥४४॥

Ca.Ci. - 13/42-44

- Consumption of sharp edged instruments such as glass, stones, wooden pieces, bones, thorns or nails in the food
- These instruments tear the walls of intestines. This mechanism will lead to the leak of contents of the intestine to the peritoneum.
- The accumulation in the peritoneum will lead distension of the abdomen leading to the manifestation of *Chidrodara*.

Lakshana of Chidrodara

- Increase of swelling below the umbilicus

- G K** 37 Baddhagudodara is a type of *Udara* where consumption of body hairs, scalp hairs along with food intake is responsible this type of the disease. This is responsible for vitiation of *Apana Vata* resulting in reverse movement of the same, leading to *Udavarta*. This is attributable to chronic intestinal obstruction.
- 38 *Chidrodara* is the type of *Udara*, where disease manifests due to the injury occurs to the intestines. This is responsible for reddish, yellowish etc discolouration of faecal matter.

- Depending upon the *Dosha* vitiation, the clinical features may vary.
- Reddish, bluish, yellowish, slimy passage of stools with cadaveric smell
- Hiccups
- Breathing discomfort
- Cough
- Thirst
- Excess micturition
- Tastelessness
- Impaired digestion
- Debility

Jalodara³⁹

नेहपीतस्य मन्दाः क्षीणस्यातिकृशस्य वा।
अत्यम्बुपानात्रष्टेऽग्नौ मारुतः क्लोनि संस्थितः॥४५॥
स्रोतःसु रुद्धमार्गेषु कफचोदकमूच्छितः।
वर्धयेतां तदेवाम्बु स्वस्थानादुदराय तौ॥४६॥
तस्य रूपाणि - अनन्नकाङ्क्षा पिपासागुदस्त्रावशूलश्वासकास
दौर्बल्यानि अपि चोदरं नानावर्णरा- जिसिरा संततमुदकपूर्णं
वृत्तिकोभसंस्पर्श भवति, एतदुदकोदरमिति विद्यात्॥४७॥

Ca.Ci. - 13/45-47

- In the person who is having *Mandagni* who has taken *Snehapana*
- The person who has excess depletion of body tissues
- Excess emaciated individual
- The person who has loss of appetite due to excess drinking of water
- Vata Dosha* gets vitiated to harm and obstruct the channels of *Kloma* which is the *Mula* of *Udakavaha Srotas*. *Udaka* and other *Dosha* gets deviated to *Koshtha* and starts accumulating in *Koshtha* to produce *Udara*. As the chief content of *Udara* in this situation is *Jala* (water), the name *Jalodara* came to this particular clinical condition.

Lakshana of Jalodara

- Tastelessness
- Thirst
- Secretion from anus
- Pricking type of pain in abdomen
- Breathing discomfort
- Cough
- Debility
- Abdominal swelling with visible veins having

³⁹ In this condition, the presence of *Jala* (water) is observed in *Udara* region. In other words, more amount of water is accumulated in peritoneal cavity.

As per *Ayurveda*, 3 stages of *Jalodara* are mentioned. *Ajatodakavastha*, *Picchavastha* and *Jatodakavastha*.

multiple colours, accumulation of water in the abdomen. On tapping the abdomen, we can observe the fluid thrill.

3 stages of Jalodara are mentioned by Charaka.

1. *Ajatodaka* stage (where no fluid accumulation in the abdomen)
2. *Piccha* stage (slimy stage where sliminess is present in the abdominal region)
3. *Jatodaka* stage (where fluid accumulation in the abdomen)

Ajatodaka Avastha of Jalodara

अजातशोथमरुणं सशब्दं नातिभारिकम्।
सदा गुडगुडायं च सिराजालगवाक्षितम्॥५५॥
नाभि विष्टभ्य पायौ तु वेगं कृत्वा प्रणश्यति।
हन्नाभिवङ्गणकटीगुदप्रत्येकशूलिनः॥५६॥
कर्कशं सृजतो वातं नातिमन्दे च पावके।
लोलस्याविरसे चास्ये मूत्रेऽल्पे संहते विषि॥५७॥
अजातोदकमित्येतैर्लिङ्गैर्विज्ञाय तत्त्वतः।

Ca.Ci. - 13/55-58

उपाक्रमेद्भिषग्दोषबलकालविशेषवित्॥५८॥

- ⊗ This is the stage of *Udara* where notable amount of fluid is not accumulated in the *Udara* region.
- ⊗ Reddish colour of abdomen
- ⊗ Gurgling sound in the abdomen
- ⊗ No heaviness of the abdomen
- ⊗ Abdominal wall is rigid with visible network of veins
- ⊗ Abnormal *Vata* will move through anus by stiffening umbilical region
- ⊗ Pricking type of pain in chest, umbilical region, groin, waist and anal region
- ⊗ Loud and harsh sound is produced while passing the flatus
- ⊗ Medium digestive fire
- ⊗ Excessive salivation and reduced micturition

Piccha stage of Jalodara:

This is the intermediate stage of *Ajatodaka* and *Jatodaka* stage of *Jalodara*. This stage is characterized by slimy state of *Udara* region.

यति; तत्र पिच्छोत्पत्तौ मण्डलमुदरं गुरु स्तिमित माकोठितमशब्दं
मृदुस्पर्शमपगतराजीकमाक्रान्तं नाभ्यामेवोपसर्पति।
ततोऽनन्तरमुदकप्रादु-

Ca.Ci. - 13/48

- ⊗ Globular swelling of the abdomen
- ⊗ Heaviness of the abdomen
- ⊗ Sensation of wet cloth covering the whole body

- Percussion sound can be heard
- Soft in touch in the abdominal region
- Visible veins which radiate from the umbilicus

Jatodaka stage of Jalodara: This is the third and last stage of *Jalodara*. In this stage, there will large amount of accumulation of *Jala* in the abdominal region. This is incurable in prognosis. The clinical features of *Jatodaka* are as follow:

यति; तत्र पिच्छोत्पत्तौ मण्डलमुदरं गुरु स्तिमितमाकोठितमशब्दं
मृदुस्पर्शमपगतराजीकमाक्रान्तं नाभ्यामेवोपसर्पति।

ततोऽन्तरमुदकप्रादु-

Ca.Ci. - 13/48

- Excessive increase in the size of abdomen due to the accumulation of fluid in abdomen
- Visible veins seen in *Picchavastha* looks to enter deep inside the abdominal region
- Excess accumulation of fluid in the abdomen can be easily examined with necessary physical examination i.e. fluid thrill.

Upadrava of Udara

तदाऽऽतुरमुपद्रवाः स्पृशन्ति - छर्यतीसारत- मक्तृष्णा-
श्वासकासहिका दौर्बल्य पार्श्वशूलरुचि- स्वरमेदमूत्र-सङ्गादयः;
तथाविधमचिकित्स्यं विद्यादिति॥49॥

Ca.Ci. - 13/49

- Vomiting⁴⁰
- Diarrhoea⁴¹
- Darkness in front of the eyes
- Thirst⁴²
- Breathing discomfort
- Cough

- Hiccups⁴³
- Debility⁴⁴
- Pain in sides of the abdomen⁴⁵
- Tastelessness
- Hoarseness of voice
- Retention of urine⁴⁶

Asadhya Lakshana of Udara

शूनाक्षं कुटिलोपस्थमुपक्लिन्नतनुत्वचम्।

बलशोणितमांसाग्निपरिक्षीणं च वर्जयेत्॥52॥

श्वयथुः सर्वममोत्थः श्वासो हिव्काऽरुचिः सतृद्र।

मूर्च्छाच्छर्दिरीती सारो निहन्त्युदरिणं नरम्॥53॥

जन्मनैवोदरं सर्वं प्रायः कृच्छ्रतमं मतम्।

बलिनस्तदजाताम्बु यत्नसाध्यं नवोत्थितम्॥54॥

Ca.Ci. - 13/52-54

- Swelling of *Indriya*⁴⁷
- Abnormal shape of genital organs
- Moist and thin skin of the individual⁴⁸
- Extreme loss of physical strength
- Extreme loss of blood in the body
- Extreme muscular atrophy in the affected individual⁴⁹
- Extreme loss of appetite⁵⁰
- Swelling of all *Marma*
- Breathing discomfort
- Hiccup⁵¹
- Tastelessness
- Thirst⁵²

- 40 This can be observed when there is backflow of the blood in the stomach. This will lead to stretch of the stomach wall leading to vomiting.
- 41 This can be observed when there is backflow of the blood into the small and large intestine and there is poor absorption from the intestines.
- 42 These symptoms can be attributable to hypovolemia as there will be shift of fluid from the body into peritoneal cavity.
- 43 These symptoms are attributable to excess accumulation of the fluid in the peritoneal cavity which compresses the diaphragm leading to the compromise in the respiratory function.
- 44 This is attributable to poor nutrient absorption and the person will have increased mobility of fat into the liver leading to the fatty degeneration and thereby the person will have appearance of debility.
- 45 Pain on the sides of the abdomen indicates the excess stretching of the abdominal wall due to the presence of excess fluid in the peritoneal cavity.
- 46 This is attributable to poor renal blood supply which occurs as a secondary manifestation of huge ascites.
- 47 Swelling of the *Indriya* such as skin is attributable to the reduced concentration of protein in the blood which may lead to generalised slow pitting type of swelling.
- 48 Moist and thin skin of the abdominal wall is attributable to huge ascites.
- 49 This is attributable to nutrient loss either carbohydrate or fat or proteins or micronutrients. This occurs due to the shift of the excess displacement of the fluid from the intravascular compartment into interstitial compartment (peritoneal cavity). Thus, other parts of the body receive minimum amount of the blood supply, thus, the person may have above said clinical features.
- 50 This is attributable to the consequence of portal hypertension. Because of portal hypertension, there will be backflow of the blood to the superior mesenteric vein and splenic vein. When there is backflow of the blood into the superior mesenteric vein, the person may have increased secretion in the intestinal lumen and if there is backflow of blood, the person may have increased secretion in the gastric lumen. In both situations, the person may develop loss of appetite.
- 51 This is attributable to the huge ascitic swelling which compresses diaphragm leading to the upward push of the lungs and thus person may present with breathing discomfort or hiccups.
- 52 This is attributable to reduced fluid in the intravascular compartment which occurs as a result of increased accumulation of fluid in the interstitial compartment, peritoneal cavity in particular. This leads to the manifestation of hypovolemia and in turn the person may develop thirst.

Altered state of consciousness⁵³

Vomiting

Diarrhoea⁵⁴

By nature all types of Jalodara from the beginning of the pathogenesis, is difficult to cure

If the Rogibala is more and the person is having Ajatodaka (absence of fluid in the abdomen) stage of the illness and Udara which is of short duration such disease is considered to be treatable with much effort.

Sadhyasadhyata of Udara

वातात्पित्तात्कफात् प्लीहः सन्निपातात्तथोदकात्।

परं परं कृच्छ्रतरमुदरं भिषगादिशेत्॥50॥

प्रायो भवत्यभावाय च्छिद्रान्नं चोदरं नृणाम्॥51॥

पक्षाह्णगुदं तूर्ध्व सर्वं जातोदकं तथा।

Ca.Ci. - 13/50-51

Vataja Udara, Pittaja Udara, Kaphaja Udara and Sannipataja Udara are considered as having increase in severity in the successive order.

All types of Udara finally lead to the manifestation of Jalodara.

Baddhagudodara⁵⁵ disease is incurable when the duration of the illness crosses 15 days.

Jalodara of any cause is considered to be Asadhya (incurable) in nature.

Chidrantra⁵⁶ is considered to be an immediately life threatening disease among the types of Udara.

Splenomegaly

Splenomegaly is the swelling of the spleen which is seen in left hypochondriac region.

This is usually examined by bimanual method of palpation. Situation of spleen, size, shape, consistency, surface, edge, tenderness, movement of spleen with respiration is examined while assessment of spleen.

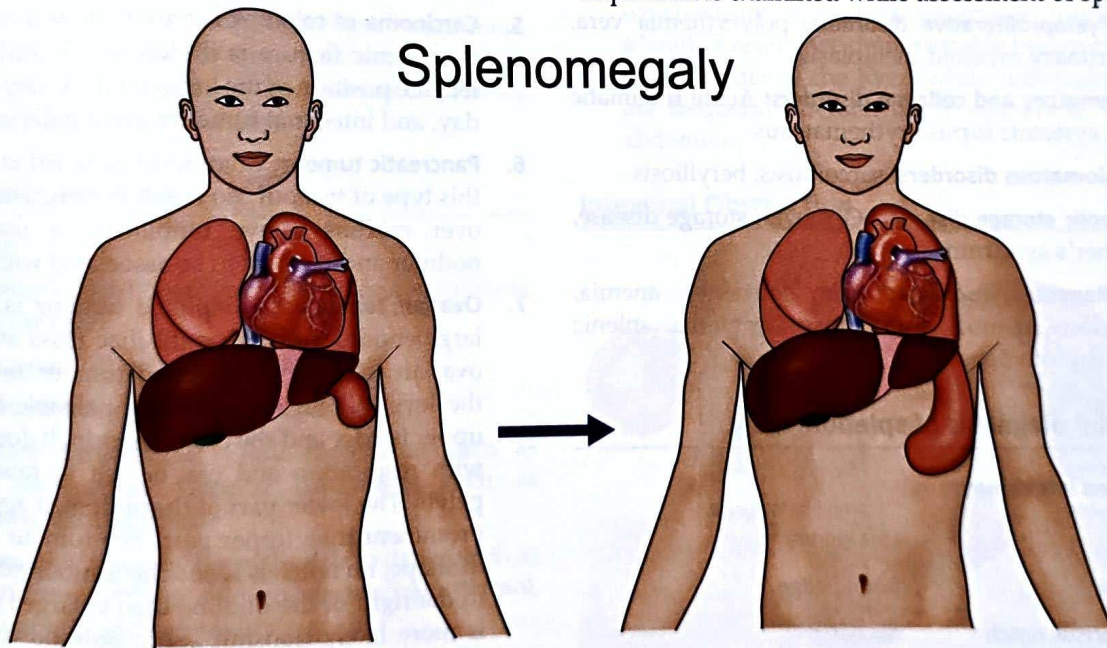


Figure 75: Clinical differentiation of normal spleen and splenomegaly

⁵³ This is attributable to hypovolemia leading to the manifestation of either hypovolemic shock or syncopal episode.

⁵⁴ This is attributable to the consequence of ascites manifestation as a result of either intestinal tuberculosis or malignant ascites depending upon the nature of progression of pathological events.

⁵⁵ This can be attributable to chronic intestinal obstruction. This may be associated with obstipation or colicky type of abdominal pain or faeculent vomiting or visible peristalsis or oliguria or tetany in as late manifestation.

⁵⁶ The term Chidrantra can be considered as injury to the abdomen.

Abdominal injuries are broadly classified into closed (blunt or crush) injuries and open (penetrating or stab) injuries.

Closed injuries: is caused by compressive force such as fall from height, blow by fist or by blunt instruments e.g. lathi. Crush injuries may occur in run over accidents or fall of heavy objects on the abdomen. Seat belt injury is also included in the group which may cause even the rupture of intestine.

Open injuries: is caused by sharp cutting instrument e.g. knife or razor or by penetrating injury e.g. bullet or missile. In this type of injury, peritoneal cavity is exposed outside and peritonitis is inevitable.

When there is internal haemorrhage, the person may have pallor, restlessness, small thread pulse, deep and sighing respiration (air hunger), subnormal temperature and collapse are seen.

Depending upon the organ injured such as liver, spleen, small intestine, large intestine, kidneys, urinary bladder etc, the person may show different clinical features which are usually life threatening if proper care is not taken.

Causes of splenomegaly

1. **Infections:**
 - a. **Bacterial:** septicemia, typhoid, infective endocarditis, tuberculosis, brucellosis
 - b. **Viral:** hepatitis, infectious mononucleosis
 - c. **Protozoal:** malaria, kala-azar
 - d. **Rickettsial:** typhus
2. **Circulatory:** congestive cardiac failure, portal hypertension, hepatic or portal vein thrombosis, splenic vein obstruction
3. **Haematological:**
 - a. **Hemolytic disorders:** hereditary spherocytosis, hereditary elliptocytosis, thalassemia, hemoglobinopathies
 - b. **Malignancy:** acute leukemia, chronic myeloid leukemia, chronic lymphocytic leukemia, lymphoma
 - c. **Myeloproliferative disorders:** polycythemia vera, primary myeloid metaplasia
4. **Inflammatory and collagen disorders:** Acute rheumatic fever, systemic lupus erythematosus
5. **Granulomatous disorders:** sarcoidosis, berylliosis
6. **Metabolic storage diseases:** Glycogen storage disease, Gaucher's syndrome
7. **Miscellaneous disorders:** Iron deficiency anemia, pernicious anemia, amyloidosis, myeloma, splenic cysts, thyrotoxicosis

Differential diagnosis of splenomegaly

1. Enlarged left kidney:

Spleen	Left kidney
Sharp edge	Round edge
A characteristic notch	No notch
Angular poles	Rounded poles
Superficial	Deep
Fingers cannot be pushed between spleen and costal margins	Fingers can be pushed between kidney and costal margin
Dull on percussion	Vertical band of colonic resonance on front
Moves freely with respiration	Movement with respiration is not so marked
Moves early on inspiration	Moves late on inspiration
Direction of movement on respiration downwards and to the right	Movement directly downward on respiration

Tendency to bulge forwards and inwards	Tendency to bulge into the loin
The loin is resonant on percussion posteriorly	No area of resonance between mass and the spine, the whole loin is dull
Cannot be pushed back into the loin	Can be pushed back into the loin

2. **Liver enlargement:** A greatly left lobe of liver may simulate splenic swelling. The enlargement of liver will be usually be tender, a notch is usually absent, the surface may be irregular.
3. **Carcinoma of stomach:** The edge of carcinomatous lump involving body of stomach is not well defined, a notch is absent, it extends more to the right than upwards.
4. **Suprarenal tumour:** The physical signs of this tumour may be closely similar to those of kidney tumour. In this occasion, it is easier to pass the fingers between the swelling and left costal margin.
5. **Carcinoma of colon:** A carcinomatous mass involving the splenic flexure is ill-defined, a notch cannot be felt, the position of the lump tends to vary from day to day, and intestinal tumours are usually conspicuous.
6. **Pancreatic tumour:** Commonly situated in midline. In this type of tumour, no notch is seen, usually crosses over midline above umbilicus, is usually more nodular and is likely to be associated with jaundice.
7. **Ovarian tumour:** The splenic tumour is sometimes large enough to reach right iliac fossa and simulate ovarian tumour. An ovarian tumour rarely attains the level of ribs, a gap being palpable between its upper border and the costal margin. It does not move with respiration and can be felt to raise from the pelvis. The lower part of the abdomen appears more prominent than upper part. The tumour is globular in shape, no notch is seen. The tumour extends more to the right of the midline than enlarged spleen and is more likely transmit aortic pulsations. A vaginal examination confirms its ovarian origin.
8. **Faecal accumulations:** The splenic flexure or adjacent part of the transverse or descending colon when loaded with faeces has been mistaken for enlarged spleen. Such a mass is however more irregular and cylindrical in shape and can be indented by the finger.

Hepatomegaly

- ① The term hepatomegaly refers to the increase in size of the liver. While discussing hepatomegaly, we need to differentiate visceroptosis of the liver or descent of liver. Visceroptosis is actually displacement of liver from its place due to multiple reasons such as massive pleural effusion or empyema of right pleural cavity etc.

Enlarged liver



Figure 76: Image of hepatomegaly

- Palpation of the liver should be done by using radial border of index finger of the right upper limb to identify the lower border of the liver and percuss border of liver. This will help to measure the size of the liver.
- Size of liver, shape, edge, surface, tenderness, consistency, movement of liver with respiration and sudden change in the size of the liver has to be noted during the examination of liver.

Causes of Hepatomegaly

1. **Infection:**
 - a. **Viral:** infective hepatitis, infectious mononucleosis, yellow fever
 - b. **Bacterial:** Tuberculosis, typhoid
 - c. **Protozoal:** Amoebiasis, malaria, kalazar
2. **Circulatory disturbances:** Local congestion such as portal obstruction; General congestion such as congestive cardiac failure
3. **Toxic hepatitis:** Chronic hepatitis due to drugs such as isoniazid, methyl dopa, chlorpromazine, paracetamol, nitrofurantoin and cytotoxic agents
4. **Reticulosis:** Leukemia, Hodgkin's disease, non Hodgkin's lymphoma, multiple myeloma
5. **Biliary obstruction:** Biliary calculi, tumour or stricture of the bile ducts
6. **Infiltrations:** Fatty infiltration, lipid storage diseases
7. **Tumours and cysts:** Benign and malignant hepatoma, benign and malignant cholangioma, secondaries, hydatid cyst
8. **Myeloid metaplasia:** Secondary carcinoma of bone, myelofibrosis, multiple myeloma
9. **Genetic abnormalities:** Sickle cell disease

Differential diagnosis of Hepatomegaly

1. **Malignancy of stomach:** malignancy of stomach

particularly involving greater curvature of the stomach. The abdominal mass, although usually palpable in the epigastrium, may also be felt in the left upper, right upper or lower quadrant of abdomen depending upon the location of tumour in the stomach. A stomach tumour is usually movable, tender and tends to display visible peristalsis even when its borders are ill-defined.

2. **Carcinoma of transverse colon:** It is characteristically hard, nodular and mobile (unless fixed by adhesions) whenever palpable, in case it causes intestinal obstruction, there is usually audible or visible peristalsis with abdominal distension or tympanitis.
3. **Kidney tumour:** A massive tumour arising from the right kidney is usually smooth and rounded, gives a tense or elastic feel to the palpating hand, occupies a lateral position and is bimanually palpable swelling.
4. **Omental mass:** A thickened or rolled-up greater omentum, secondary to tuberculous peritonitis, usually presents a tender, sausage shaped lump, which hardly moves with respiration; there is usually a band of resonance demonstrable between it and the lower border of the liver, while tuberculous glands are frequently demonstrable elsewhere within the abdomen.

Intestinal Obstruction

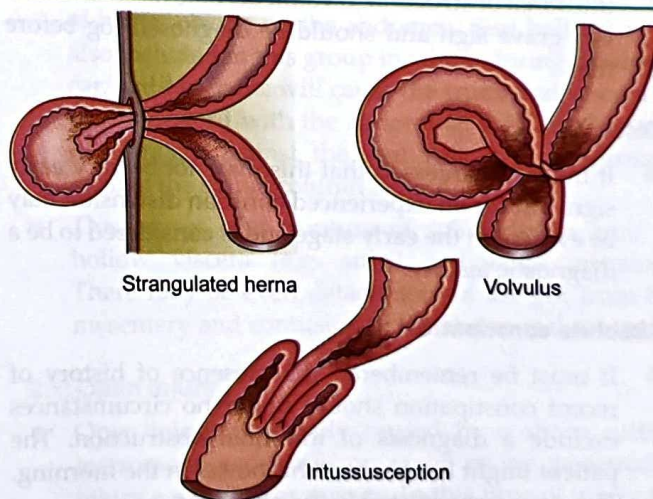


Figure 77: Images of different varieties of intestinal obstruction

Intestinal obstruction is another major condition where in we can observe the distension of the abdomen. This could be acute or chronic. The acute intestinal obstruction is considered to be a case of acute abdomen. The mortality rate from acute intestinal obstruction rises with each passing hour from the onset of the disease, so early diagnosis is very much essential. The three main clinical manifestations of acute intestinal obstruction are intestinal colic, vomiting and distension of the abdomen. The absolute constipation (obstipation) is considered to be one of the cardinal features should be excluded as it

important symptom.

- Generalized appearance of sepsis and systemic upset become apparent. On examination, epigastric and flank tenderness can be elicited. The abdomen may be slightly distended and on auscultation there will be diminished peristaltic sound.

Small intestine

- Rupture of small intestine may also occur from blunt injury or penetrating injury to the abdomen. In case of penetrating injury the symptoms and signs are more or less like penetration of peptic ulcer. In case of blunt injury, the symptoms and signs may not appear early and this delay is due to appearance of traumatic necrosis involving the parts of the injured intestine.
- The fixed parts of the small intestine are more vulnerable to blunt injury particularly when the blow impinges these parts against the vertebral column. It is the first two feet of jejunum and last two feet of terminal ileum which are most vulnerable to closed traumatic rupture. Early diagnosis is highly important.
- The time interval between the perforation and development of peritonitis depends on the size of the rupture, whether the perforation occurs into the free peritoneal cavity and on the character of the intestinal contents. Pointing test is an important sign in identifying rupture of intestine.
- The patient is asked to point which is the most painful area or where the pain started. This often indicates the site of perforation. London sign indicates pattern of bruising of the skin (i.e. an imprint of clothing or a seat belt noted on the skin) with the type of crushing force applied to the abdomen against the vertebral column.
- Local tenderness is highly important as it often indicates the site of rupture. Rebound tenderness means peritonitis has set in due to rupture of small intestine.

Large intestine

- Rupture of large intestine may be intraperitoneal or extraperitoneal. Rupture of ascending and descending colon may be intra or extraperitoneal, whereas rupture of transverse and sigmoid colon is often intraperitoneal.
- Intraperitoneal rupture will lead to severe peritonitis as the contents of large intestine are highly infective. The condition is lethal. Radiography may be helpful in detecting large pneumoperitoneum suggests escape from the predominantly gas containing large intestine.
- Extraperitoneal injury will lead to spreading cellulitis and surgical emphysema in the loins.

Kidney

- Renal injuries can be classified into slight, severe and critical. Slight injuries comprise those where the parenchyma is damaged without rupture of the capsule or extension of the laceration into the renal pelvis or calyx.
- This also includes a contusion of the cortex of the kidney without tear of the capsule and this produces a subcapsular haematoma. This condition does not haematuria, but slight tenderness at the renal angle can be elicited. Severe injuries are those where the capsule is broken, renal pelvis or calyx is distorted.
- This produces haematuria or a mass in the loin from a perirenal haematoma. There may be clot colic. There may be leakage of urine in the retroperitoneal tissue. Perinephric haematoma is suspected when there is flattening of the normal curvature of the loin. In many cases of renal injuries, there will be generalized abdominal distension (meteorism) which is caused by retroperitoneal haematoma pressing on the splanchnic nerves. One must continue to examine the urine for haematuria both macroscopic and microscopic.
- If haematuria gradually ceases, it is a good sign but the patient should be kept at rest for a few days or more as such cessation of haematuria may be due to occlusion of ureter by blood clot. A critical injury is such when the kidney is shattered or there is a tear in the renal artery or one of its branches. The patient rarely survives after this type of injury.
- Delayed rupture is very rare, though is occasionally seen in renal injury. A patient who after injury did not reveal any sign of kidney injury suddenly suffers from profuse haematuria usually between 3rd and 5th days of accidents. This usually occurs due to some movement which dislodges the clot into the renal pelvis. So rest in bed is extremely important even when minimum injury to kidney is suspected.

Urinary bladder

- This may be injured intraperitoneally or extraperitoneally. The rupture is extraperitoneal in 90% of the cases. Intraperitoneal rupture can only occur when someone is drunk so that his abdominal musculature remains relaxed during the blow and the bladder is full.
- Symptoms of ruptured bladder are usually marked due to multiple injuries and shock. A question must be asked 'Has he passed the urine since the accident?'
- Extraperitoneal rupture:** It is usually associated with fractured pelvis. This is ascertained by compression test by compressing the pelvis laterally inwards or by distraction test by distracting the pelvis. Fracture is indicated by pain. Extravasation occurs into the

to the hollow viscus. Internal haemorrhage produces certain general signs which are common for injury to any viscus inside the abdomen. But the local signs depend on the injury to the concerned viscus.

- In case of internal haemorrhage, increasing pallor, restlessness, small thready pulse, deep and sighing respiration (air hunger), subnormal temperature and collapse are the general signs. Specific signs are discussed below:

Liver

- In case of injury to the lower part of the chest on the right side one may suspect of ruptured liver. Liver injury may occur as a result of penetrating wound by stabbing or bullet injury.
- Right lobe is more commonly injured (5:1) than the left lobe as it is less mobile and large. Patients complain of pain in the right upper quadrant of the abdomen.
- Occasionally, when the central part of liver is ruptured, bleeding occurs into the large radicles of the biliary tree so that liquid blood is carried along the bile passages into the duodenum and there is haemetemesis.
- Examine for tenderness and rigidity in the right hypochondriac region. Look for increased area of dullness, shifting dullness and silent abdomen.

Spleen

- Injury to spleen is usually caused by closed (blunt) injury left upper quadrant injury and only occasionally it may be caused by penetrating injury to the lower costal margin.
- This is more common in tropical countries where spleen is diseased by malaria. Rupture of spleen manifests itself in one of the three following types –
 1. The patient succumbs rapidly.
 2. There are immediate signs of rupture (commonest type).
 3. Delayed type.
- 1. **The patient succumbs rapidly** – This occurs due to tearing of the splenic vessels and complete avulsion of the spleen from its pedicle. This leads to catastrophic internal haemorrhage and becomes fatal within minutes.
- 2. **There are immediate signs of rupture (commonest type)** – This is the commonest group seen in present clinical practice or surgical practice. There are signs of initial shock when it may not be possible to state precisely which organ has been damaged. Gradually this initial shock is recovered and the signs of intra-abdominal bleeding caused by ruptured spleen will become evident. After moderate intra-abdominal haemorrhage, adequate clotting occurs to control the

haemorrhage temporarily. Thus when the patient is first seen the general signs (which are discussed earlier in this chapter) may not be that alarming, but the local signs become more important to come to a diagnosis.

- **Local signs:** There are tenderness and muscle guard over the left upper quadrant.
 - Abdominal distension (meteorism) gradually appears about 3 to 4 hours after the accident. It is due to paralytic ileus.
 - **Kehr's sign:** Pain is referred to the left shoulder due to irritation of the left half of the diaphragm by splenic blood. There may be hyperaesthesia in the area of left shoulder. This sign may be demonstrated by elevating the foot of the bed for 15 minutes by which time blood will accumulate below the left cupula of the diaphragm.
 - **Presence of shifting dullness in the flanks is fairly common. Ballen's sign:** Persistent dullness on the left side of the abdomen due to early coagulation of splenic blood. Even at this stage, shifting dullness is present on the right side.
 - Rectal examination may reveal tenderness and sometimes a soft swelling due to presence of blood or clot in the rectovesical pouch.
 - Saegesser's splenic point is the point in the lower part of the posterior triangle of the neck between the left sternocleidomastoid and the scalenus medius muscles above the clavicle. Application of pressure on this point will give rise to severe pain.
3. **Delayed type:** In this type, following the accident, there is period of comparative freedom of symptoms for a few days (15 days or more). After which the patient is often readmitted with well marked signs of internal haemorrhage.

Mesentery

- Both closed and open injury can in mesenteric laceration. As expanding haematoma which compresses the arterial arcade may threaten the visibility of segment of intestine.
- So after injury to the abdomen when tenderness and rigidity persist and the patient's condition gradually deteriorates in spite of good resuscitative treatment one should immediately consider of opening the abdomen.

Duodenum

- The intraperitoneal rupture gives rise to the similar clinical picture as perforation of peptic ulcer. Retroperitoneal ruptures are more difficult to detect.
- Symptoms develop a few hours or even a day after injury. Severe pain in the epigastrium, in the back associated with intractable vomiting is the most

elderly, the onset is gradual (so it is often said to be chronic). Pain is not very prominent. If present it is complained of in the hypogastrium.

• Vomiting is a very late feature and it may not be present altogether. Distension is more marked on the flanks.

• Careful palpation of the right iliac fossa may reveal the caecum to be distended and may harden under the examining finger. Rectal examination may reveal an annular growth.

Nature of obstruction

• Intestinal obstruction may be simple or strangulated and dynamic or paralytic. In simple obstruction, there will be intestinal colic.

• Pulse, respiration, temperature and blood pressure remain unaffected. On palpation, muscle rigidity and rebound tenderness are conspicuous by their absence.

• In strangulation, however, there is no complete remission of pain between colics. So colicky pain of simple obstruction is replaced by a continuous pain.

• Pulse becomes rapid, temperature may be elevated, there may be tachypnoea and blood pressure falls from the beginning.

• Presence of muscle rigidity and rebound tenderness immediately make the diagnosis. Two hours gastric suction if fail to relieve pain indicates intestinal strangulation.

• Though external strangulation is easier to diagnose by tense, tender and irreducible swelling without any impulse on coughing, yet internal strangulation is not so easy to diagnose and must be looked for.

Paralytic ileus

May occur in the early post-operative period after an abdominal operation or as a complication of diffuse peritonitis. It is differentiated from dynamic intestinal obstruction by

• History of abdominal operation or features of peritonitis are present

• Gradual distension of the abdomen without visible peristalsis

• Absence of colicky pain

• Silent abdomen on auscultation

Cause of obstruction

• In determining the cause of obstruction, age of the patient should be taken into consideration first.

In the new born, cause of obstruction is congenital malformation e.g. congenital atresia of oesophagus, congenital pyloric stenosis or meconium ileus, congenital atresia of duodenum or ileum.

• In infancy, intussusception and obstruction due to worms.

• In adolescence, besides, intussusception band or adhesion resulting from Meckel's diverticulum or local peritonitis due to appendicitis, tubercular peritonitis etc.

• In adults, obstruction by band, volvulus or growth should be considered whereas in the old, carcinoma of the colon is the usual cause. A strangulated hernia can occur at any age.

Injury to the abdomen

Abdominal injuries can be broadly classified into two groups –

1. Closed (brush or crush) injury and
2. Open (penetrating or stab) injury.

1. Closed injury

• Closed injury is caused by compression force such as falling from height, blow by a fist or by blunt instruments e.g. lathi.

• Crush injuries may occur in run-over accidents or fall of heavy objects on the abdomen. Seat belt injury is also included in this group in which during driving a car, sudden brake will cause the trunk and viscera to move forward with the abdominal wall and become decelerated against the seat belt and compressed against the spinal column behind.

• This causes even rupture of solid viscera and/ or hollow viscera (e.g. small and large intestine). There may be even detachment of the gut from the mesentery and contusion of the abdominal contents.

2. Open injury

• Open injury is usually caused by a sharp cutting instrument e.g. knife or razor or by penetrating injury e.g. bullet or missile. In this type of injury the peritoneal cavity is exposed outside and peritonitis is almost inevitable.

• The intra-abdominal organ which may be injured by such injury depends on the site of this penetrating wound. As for example, stab injury to the right upper quadrant of the abdomen may injure the liver.

• The main concern of abdominal injury is injury to the viscera inside the abdomen. This often causes internal **hemorrhage**⁵⁷ and/or peritonitis from injury

Su.Chi. – 2/51

57 तत्रान्तर्लोहितं पाण्डुं शीतपादकराननम्॥ शीतोच्छ्वासं रक्तनेत्रं मानद्धं च विवर्जयेत्॥
When there is internal haemorrhage, yellowish white discolouration of the body, coldness of the palms, soles and oral cavity, cold expiration, reddish discolouration of the eyes, distension of the abdomen are the features as per Sushruta.

takes no less than 24 hours to develop the symptom. The following are different forms of intestinal obstruction.

Intestinal colic

- When obstruction lies in the jejunum or high in the ileum the colics appear in waves at intervals of 3-5 minutes. The pain lasts for about 30 seconds.
- The interval becomes longer in obstruction of terminal ileum where it is about 8-10 minutes. The site of pain will give an indication as to the site of obstruction e.g. small intestinal cramps are referred to the epigastrium or umbilical region whereas colonic cramps to the lower abdomen (hypogastrium).

Vomiting

- This is a fairly constant symptom. The frequency of vomiting depends on the site of obstruction. In upper small bowel obstruction vomiting is much more frequent than obstruction in the terminal part of the small intestine.
- At first the vomitus is the gastric content i.e. ingested food particles and fluid, next it is the content of the duodenum, which is predominantly bile and lastly it is the content of the small intestine above the site of obstruction.
- Virtually it takes about 3 or 4 days in complete intestinal obstruction to vomit become faeculent. This is a grave sign and should be diagnosed long before this.

Distension

- It must be confessed that this may not be very early sign. But for an experienced clinician distension may be evident in the early stage and is considered to be a diagnostic feature.

Absolute constipation

- It must be remembered that absence of history of recent constipation should under no circumstances exclude a diagnosis of intestinal obstruction. The patient might have moved his bowel in the morning. The symptoms started in the afternoon.
- For constipation to arrive as a significant sign one has to wait for another 24 hours. This only makes the diagnosis late in case one includes constipation to be one of the four major signs of this condition. Again absolute constipation appears later in the process as it means failure of passage of both faeces and flatus.
- To add to more infusion intestinal obstruction may be accompanied by diarrhoea. Mesenteric vascular occlusion, Richter's hernia and pelvic abscess associated with obstruction by adhesion will produce diarrhoea instead of constipation.

Dehydration

- The higher is the obstruction more will be dehydration due to tremendous loss of water and electrolytes. To the contrary obstruction of the terminal ileum is associated with relatively small amount of loss of fluid and electrolytes as the secretions of gastro intestinal tract are reabsorbed. As a matter of fact dehydration appears late in obstruction of the terminal ileum but the distension becomes pronounced whereas in obstruction of the jejunum dehydration appears earlier but distension is much less.
- On examination the patient looks anxious and restless. In the early stage, pulse, respiration and temperature show no deviation from normal. Inspection rarely shows distension in the early stage, but a little fullness cannot be missed by an experienced clinician. Distension either central (small intestinal obstruction) or peripheral (large intestinal obstruction) or regional (volvulus of sigmoid colon etc).
- Visible peristalsis can be another sign which indicates intestinal obstruction. We have to inspect the different hernial sites – inguinal, femoral, umbilical area. On palpation, we can observe muscle rigidity and rebound tenderness which suggest intestinal strangulation. Percussion is of little value. Auscultation is very important technique, but often missed. Presence of borborygmi may indicate intestinal colic and silent abdomen indicates paralytic ileus.
- Straight X ray of the abdomen is very diagnostic. It shows gas distended loops of the intestine and multiple fluid levels.
- After diagnosing the case of intestinal obstruction, we need to identify the site, nature and cause of obstruction.

Site of obstruction

- In high intestinal obstruction, vomiting is early and profuse. The vomitus consists of gastric fluid in the beginning and bile later on. Faeculent vomiting is rarely seen. The patient becomes dehydrated very soon. The urine is scanty.
- Though distension is not marked feature, yet fullness of the epigastrium is often noticed. In this category, acute dilatation of the stomach and pyloric stenosis may be included. In low small intestinal obstruction, the onset is sudden. Pain is situated over the umbilical region.
- The vomiting is present but it is neither early nor profuse. In early stage, the vomitus consists of gastric content later on bile and becomes faeculent in the last stage. Distension is marked and is central, visible peristalsis is in the form of ladder pattern, is often noticed.
- In large intestinal obstruction, the patient is usually

prevesical space. After a few hours, there will be increasing tenderness over the lower abdomen and the pulse rate will rise.

- ⊗ These factors in association with failure to pass urine and no evidence of bladder distension will confirm the diagnosis.
- ⊗ This condition is often confused with rupture of membranous part of urethra. This is differentiated by rectal examination in which the prostate may not be palpable as it is displaced upwards in rupture of membranous part of urethra (Vermooten's sign) and some amount of blood will definitely leak from the urethral meatus in the latter condition.
- ⊗ **Intraperitoneal rupture:** In this condition, the physical findings are those of peritonitis. Patient complains of agonizing pain and is often accompanied by severe shock.
- ⊗ There will be varying degree of abdominal rigidity and a few hours later abdomen becomes obviously distended.
- ⊗ Though the patient has not passed urine he does not show any intention whatsoever to do so. There is no dullness just above the pubis as the bladder is not distended.
- ⊗ But tenderness will be present in the hypogastrium. Shifting dullness is usually present. Rectal examination reveals bulged rectovesical pouch.

Ascites

- ⊗ The term ascites refers to the accumulation of free fluid in the peritoneal cavity. Over 1,500 ml of fluid

should be collected in the peritoneal cavity before its presence can be felt by physical examination.

- ⊗ This is usually because of the large size of abdomen and its many recesses. A symmetrical bulging of the flanks of abdomen in recumbent posture and bulging of the lower abdomen in erect posture indicate the diagnosis of ascites. (see Figure 78)

Diagnosis of the ascites

It is done by

1. Fluid thrill test
2. Shifting dullness
3. Horseshoe shaped dullness
4. Puddle sign in case of minimal accumulation of ascitic fluid in peritoneal cavity as little as 120 ml peritoneal fluid.

Causes of ascites

- ⊗ **Ascites not associated with peritoneal disease**

➤ Intrahepatic sinusoidal portal hypertension

1. Cirrhosis of liver
2. Acute liver failure
3. Massive liver metastasis

➤ Extrahepatic sinusoidal portal hypertension

1. Congestive cardiac failure
2. Constrictive pericarditis
3. Budd Chiari syndrome

➤ Hypoalbuminemia

1. Nephrotic syndrome

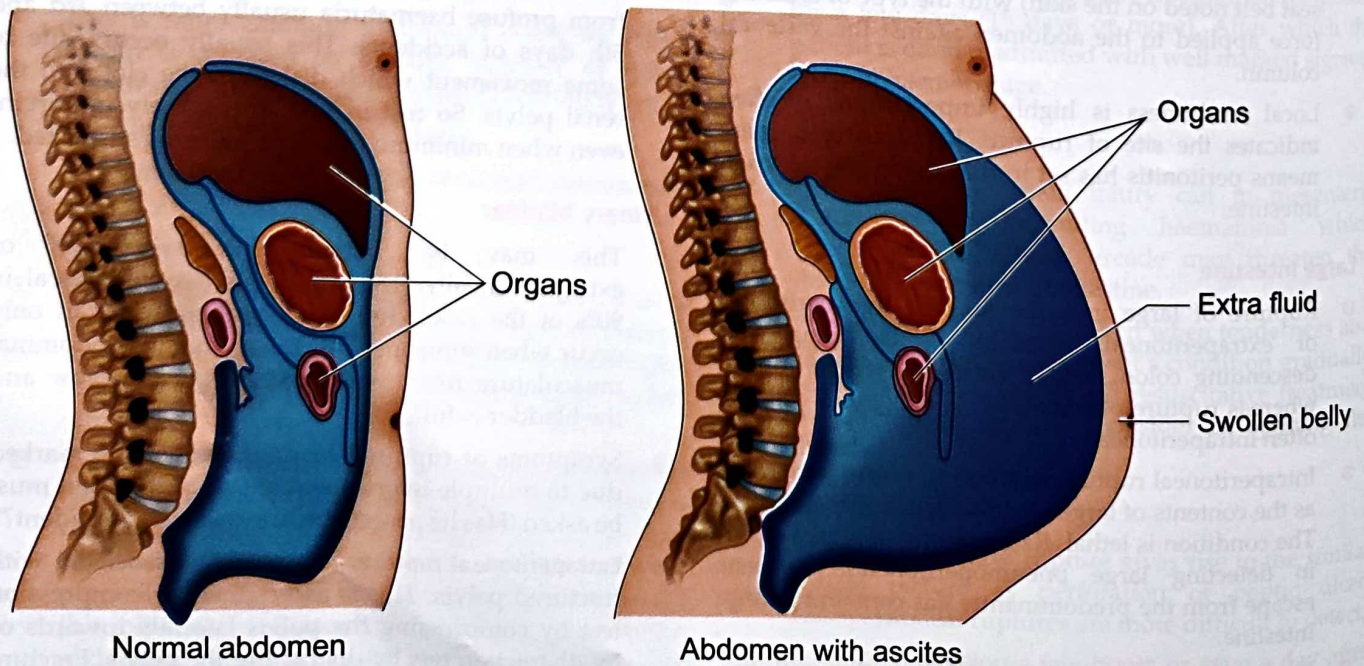


Figure 78: Image of normal abdomen and abdomen with ascites

2. Protein losing enteropathy

➤ **Miscellaneous disorders**

1. Myxoedema
2. Ovarian disease
3. Pancreatic ascites
4. Bile ascites
5. Chylous ascites
6. Nephrogenic ascites

➤ **Ascites caused by primary peritoneal disease**

- Malignant ascites
- Tubercular ascites
- Fungal and parasitic peritonitis
- Sarcoidosis
- Vasculitis

➤ **Miscellaneous peritoneal diseases**

1. Eosinophilic gastroenteritis
2. Whipple's disease
3. Endometriosis

Mechanism of ascites

The mechanism of production of ascites depends greatly on the etiological factor or factors present. It is possible that more than one pathological process may take place in an individual who is presented with ascites.

1. **Inflammation** – Irritation of the peritoneum by inflammation leads to increased permeability of capillaries and transudation of fluid into the peritoneal cavity and reabsorption of intraperitoneal fluid is reduced either because of oedema of the peritoneal membrane or because of increased protein concentration in ascitic fluid.
2. **Peripheral arterial vasodilatation** – This is the ideal mechanism occurs in the patients of cirrhosis of liver. Due to the cirrhosis of liver, there will be the commencement of portal hypertension which occurs as a backflow of the blood from the liver into portal vein. This in turn leads to the backflow of blood into splenic vein and superior mesenteric vein

thus the patient may present with loss of appetite, hematemesis due to oesophageal varices, bleeding per rectum owing to internal hemorrhoids.

3. **Lymphatic obstruction** – Involvement of the mesenteric lymphatics, receptaculum chyli or thoracic duct leads to the chylous ascites as a result of blockage of lymphatic drainage.
4. **Ruptured viscus** – Rupture of an intra-peritoneal organ by causing an outpouring of blood, cystic fluid or contaminated material from the colon results in collection of fluid within the peritoneal cavity.

Differential diagnosis of ascites

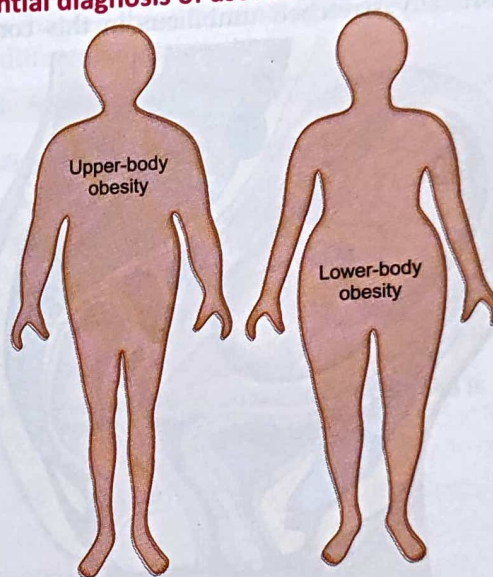


Figure 79: Apple and pear shaped obesity

1. **Obesity:** The protrusion of abdomen in obesity is symmetrical and out of proportion to the rest of the body; the navel is depressed and the transverse cutaneous folds of the abdominal wall exaggerated. In extreme obesity, the fat may hang in front like an apron with protrusion of the flanks and suprapubic region. In obesity, there is no shifting dullness or fluid wave. (see Figure 79)
2. **Gaseous distension:** There is no bulging of the flanks, the abdomen will have tympanic note all over the abdomen. (see Figure 80)

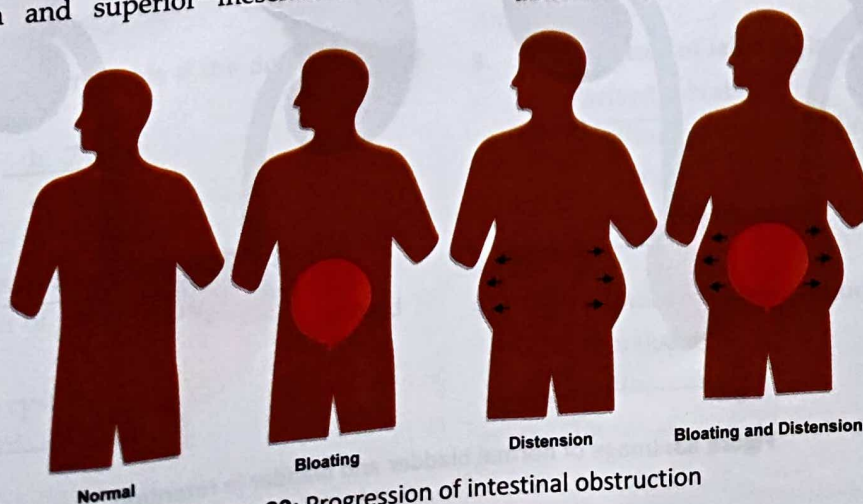


Figure 80: Progression of intestinal obstruction

3. **Pregnancy:** The gravid uterus usually causes symmetrical bulging of the abdomen with anteroposterior diameter of the abdomen markedly increased and the maximum girth of the abdomen below the umbilicus. In ascites, maximal bulging is towards the flanks of abdomen and maximum girth of abdomen is above umbilicus. In pregnancy, dullness on percussion is centrally located, with tympany at the flanks. Changes in posture produce little or no change in percussion note. (see Figure 81)
4. **Ovarian cyst:** The lady with ovarian cyst will have generalized abdominal swelling. The maximum girth of the abdomen is below the umbilicus. There will be vertically stretched umbilicus in this condition,

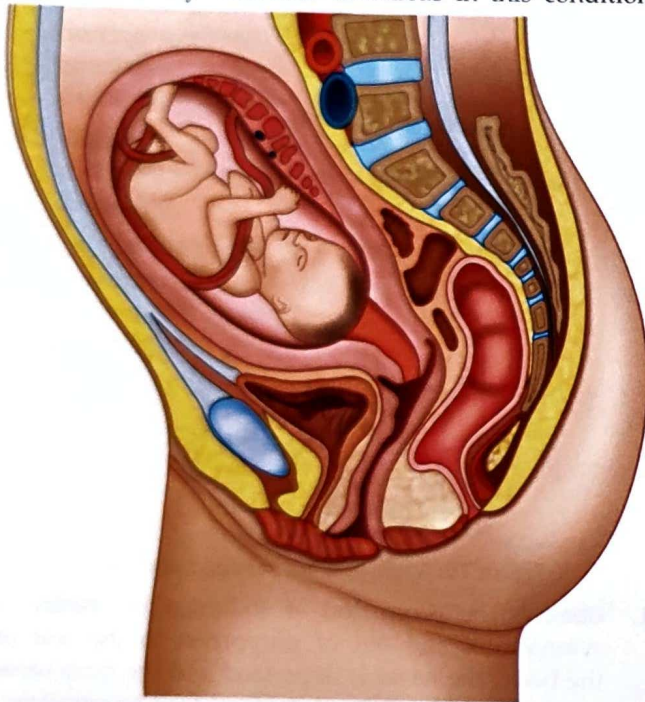
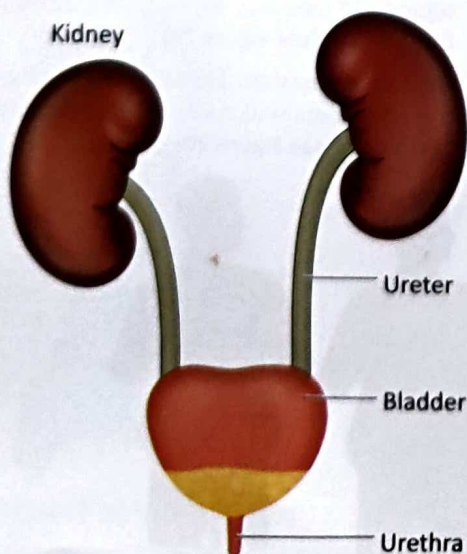


Figure 81: Abdominal swelling in pregnancy

Healthy Bladder

whereas in ascites, there will transversely stretched umbilicus is seen which is also termed as smiling umbilicus. Outline of upper border of the of dullness can be visible during respirations and this sign is not seen in the patient of ascites. (see Figure 82)

5. **Visceroptosis:** In generalized visceroptosis, the abdominal wall has a tendency to bulge laterally or in the flanks in the recumbent posture, in hypogastric region in upright posture.
6. **Splenomegaly:** Swelling more commonly seen in left hypochondriac region where we can observe the notch in the anterior border of the spleen in huge splenomegaly.
7. **Distended bladder:** A massive overdistension of the urinary bladder leads to the swelling of the suprapubic region. In this condition, the direction of umbilicus is towards upwards. There will be dullness on percussion. Swelling is localised swelling. (see Figure 83)

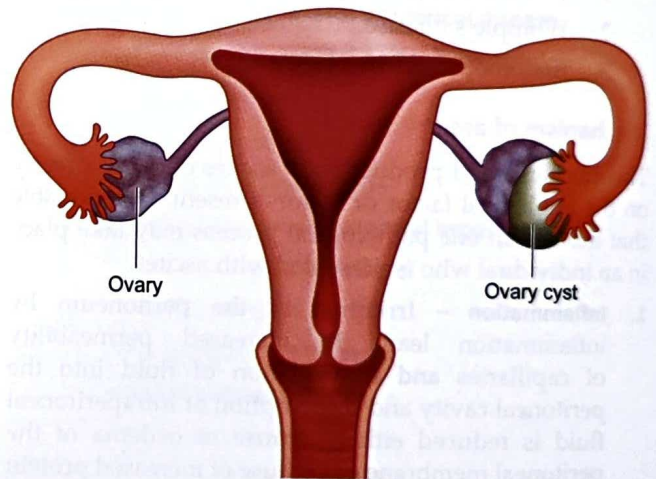


Figure 82: Ovarian cyst

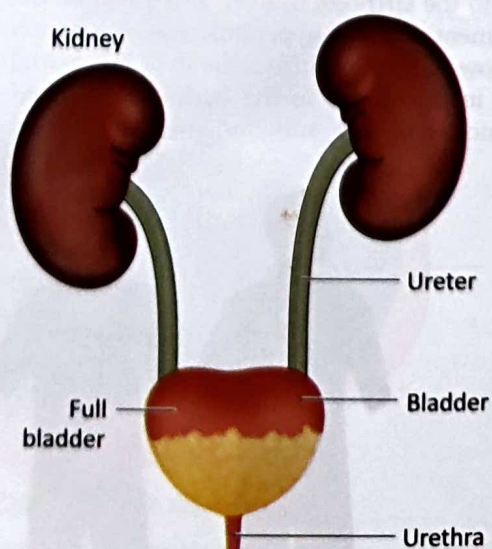
Bladder in Retention

Figure 83: Image of normal bladder and bladder in retention

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उदरोत्सेधसाधम्योदुदर

- As far as Pratyatma Lakshana of Udara is concerned, any general swelling of the abdominal region is considered as Udara. This is different from Gulma as latter disease is a localised swelling in the abdominal region.
- Mandagni or Agnimandya is considered to be the major aetiological factor in the manifestation of Udara. Apart from this, indulgence in Papakarma (socially unacceptable activities) and excess vitiation of Dosha in the diseased individual are the usual causative factors of the Udara manifestation.
- As far as Sankhya Samprapti of Udara is considered, 8 types of Udara are mentioned – Vataja, Pittaja, Kaphaja, Sannipataja, Yakriddalyudara, Baddhagudodara, Kshatodara and Jalodara. Among these, Jalodara is considered to be Asadhya in nature by default.
- Hepatomegaly is considered as the swelling of the liver. Splenomegaly refers to the swelling of the spleen. Ascites refers to the accumulation of fluid in the peritoneal cavity. In this chapter, these terminologies are explained along with the probable causes, clinical features and examination and differential diagnosis.

Self-Assessment: Questions for Review

Long essay questions

- Explain Udara in detail
- Describe the Nidana Panchaka of Udara in detail
- Explain Nidana, Samprapti, Purvarupa, Rupa, Pratyatma Lakshana, Bheda, Sadhyasadhyata of Udara.

Short essay questions

- Explain the stages of Jalodara
- Explain Yakriddalyudara
- Explain the Samprapti Ghataka of Udara
- Explain Sadhyasadhyata of Udara
- Explain ascites in detail
- Explain hepatomegaly in detail

MCQs

- Which of the following is used to test minimal amount of ascites?
 - Fluid thrill test;
 - Shifting dullness;
 - Horse shoe shaped dullness;
 - Puddle sign

Answer: (d)

- Baddhagudodara is not treatable if the duration of the illness is more than
 - 1 day;
 - 7 days;
 - 15 days;
 - 4 days

Answer: (c)

- Bulging in the flanks of the abdomen is suggestive of
 - Third trimester of pregnancy;
 - Ascites;
 - Large ovarian cyst;
 - Umbilical hernia

Answer: (b)

- Smiling umbilicus is specifically seen in
 - Ascites;
 - Large ovarian cyst;
 - Umbilical hernia;
 - Epigastric hernia

Answer: (a)

- Sankhya Samprapti of Udara as per Charaka is
 - 5;
 - 8;
 - 10;
 - 12

Answer: (b)

- Parisravi Udara is another name of
 - Sannipataja Udara;
 - Jalodara;
 - Baddhagudodara;
 - Chidrodera

Answer: (d)

- Swelling below the Nabhi region is observed in
 - Baddhagudodara;
 - Jalodara;
 - Parisravi Udara;
 - Plihodara

Answer: (c)

- The symptom of Jalodara is
 - Parivritta Nabhi;
 - Swelling between Hridaya and Nabhi;
 - Swelling below Nabhi;
 - None of the above

Answer: (a)

- The symptom of Asadhya Jalodara is/are
 - Shunaksha;
 - Kutilopastha;
 - Tanutvak

- d. All of the above

Answer: (d)

10. Which of the following test is done to determine the fluid in the peritoneal cavity?

- Puddle sign;
- Femoral nerve stretch test;
- Murphy's sign;
- Cullen's sign

Answer: (a)

11. Vertically stretched umbilicus is the sign of which of the following?

- Ascites;
- Large ovarian cyst;
- Splenomegaly;
- Intestinal obstruction

Answer: (b)

12. The minimum amount of fluid present in peritoneum to confirm the diagnosis of ascites is

- 500 ml;
- 50 ml;
- 120 ml;
- 1,000 ml

Answer: (c)

13. Which of the following is not responsible for the manifestation of splenomegaly?

- Cirrhosis of liver;
- Congestive cardiac failure;
- Malaria;
- Typhoid

Answer: (d)

14. Which of the following sign is absent in the patient of ascites?

- Wide stretch marks on the lower abdomen;
- Transversely stretched umbilicus;
- Flanks stretched laterally;
- Cullen's sign

Answer: (d)

15. Which of the following is not showing colicky abdominal pain?

- Blocked gall stone in the common bile duct;
- Impacted renal stone in the renal pelvis;

- Impaction of small part of faeculant material to the appendix;

- Ascites

Answer: (d)

Chapter resources

- https://www.researchgate.net/figure/Generalized-distension-of-the-abdomen-is-evident_fig1_45952464
- Charaka Samhita, Chikitsa Sthana – 13th chapter – Udana Chikitsa Adhyaya
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